Chapter 6, Problem 68P

Problem

A 6-V dc voltage is applied to an integrator with $R = 50\ \text{k}\Omega$, $C = 100\ \mu\text{F}$ at $t = 0$. How long will it take for the op amp to saturate if the saturation voltages are +12 V and −12 V? Assume that the initial capacitor voltage was zero.

Step-by-step solution

Step 1 of 3

Refer to section 6.6 in the textbook.

Consider the following integrator circuit diagram as shown in Figure 1.

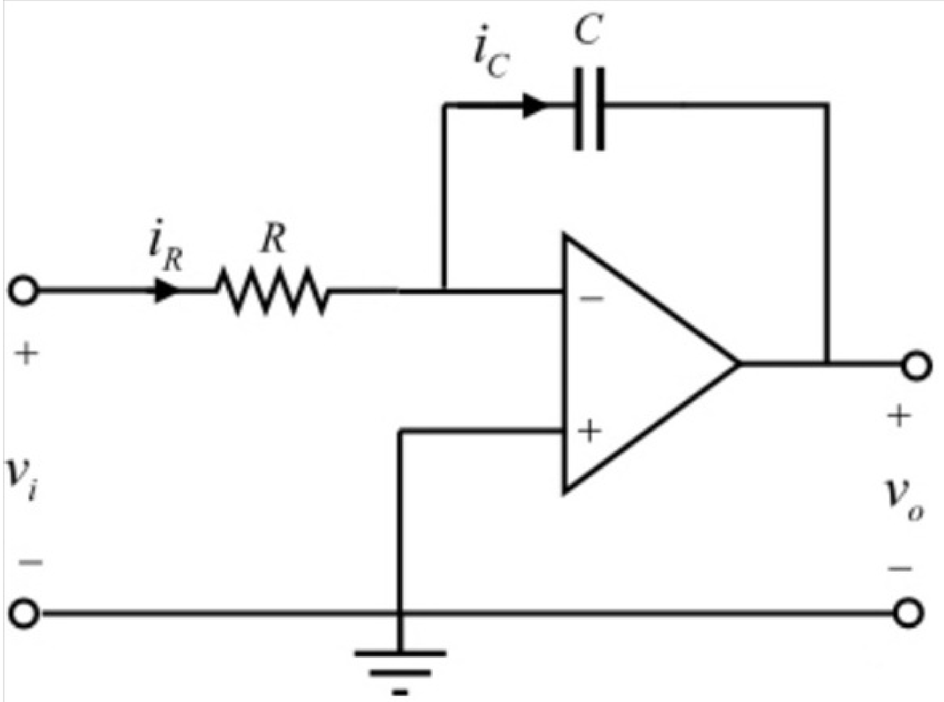

Figure 1

Step 2 of 3

Refer to equation 6.35 in the textbook.

Consider the expression for the output voltage of the integrator.

$$v_o = -\frac{1}{RC}\int_0^t v_i \, dt + v(0)$$

Substitute $50\ \text{k}\Omega$ for R, $100\ \mu\text{F}$ for C and $6\ \text{V}$ for v_i.

Assume the initial capacitor voltage is zero.

$$\begin{aligned} v_o &= -\frac{1}{(50\times 10^3)(100\times 10^{-6})}\int_0^t 6\, dt + 0 \\ &= -\frac{1}{5}\left(\left[6t\right]_0^t\right) \\ &= -\frac{1}{5}(6t) \\ &= -1.2t\ \text{V} \end{aligned}$$

$$v_o = -1.2t\ \text{V}$$

Thus, the output voltage is $v_o = -1.2t\ \text{V}$.

Step 3 of 3

Consider the Op-amp saturate at voltage $\pm 12\ \text{V}$.

$$v_o = \pm 12\ \text{V}$$

Therefore, on equating the output voltage with the saturation voltage,

$$-12 = -1.2t$$

$$t = \frac{12}{1.2}$$

$$t = 10\,\text{sec}$$

Therefore, the time taken by the op-amp to saturate at ± 12V is $\boxed{10\ \text{seconds}}$.